Generative Classifiers as Trustworthy Alternative in Tactile Classification

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Methods

Fig. 1, 36 Materials for Classification [2]

Multipurpose robots are increasingly automating industrial tasks, thanks to their enhanced ability to understand and interact with their environments [1]. A critical part of being able to appropriately react to their environment, and safely handle objects, is to be able to identify them through visual classification. Robots are already able to classify learned materials with astonishing performance [2]. A big downside of many neural networks when it comes to classification is the measure of quantifying uncertainty and explainability of any made decisions, even more in critical situations (e.g. manufacturing, interacting with humans). To address these challenges, we propose a novel architecture **TactINN**, aiming to maintain state of the art accuracy when it comes to material classification, whilst also enabling the model to recognize, when confronted with out-of-distribution data (i.e., materials it was not trained on). Additionally, our approach utilizes explainable feature maps to increase transparency of its decisions. Specifically, we adapt a generative classifier based on an invertible neural network originally designed for image classification to process tactile data [3]. In this work, we focus on the 1-sweep case, as described in [2].

Our experimental approach consists of **two main parts**:

- first, we **replicate the state-of-the-art classification model**
- second, we **adapt the generative classifier for tactile data** and compare its performance.

Additionally, we conduct a **sensitivity study** on the state-of-the-art model **to examine the relationship between sample length and classification accuracy**, aiming to improve robustness in real-world scenarios..

Video Link to Original Tactile Classification Task

State-of-the-Art Replication

Our TactNet-II

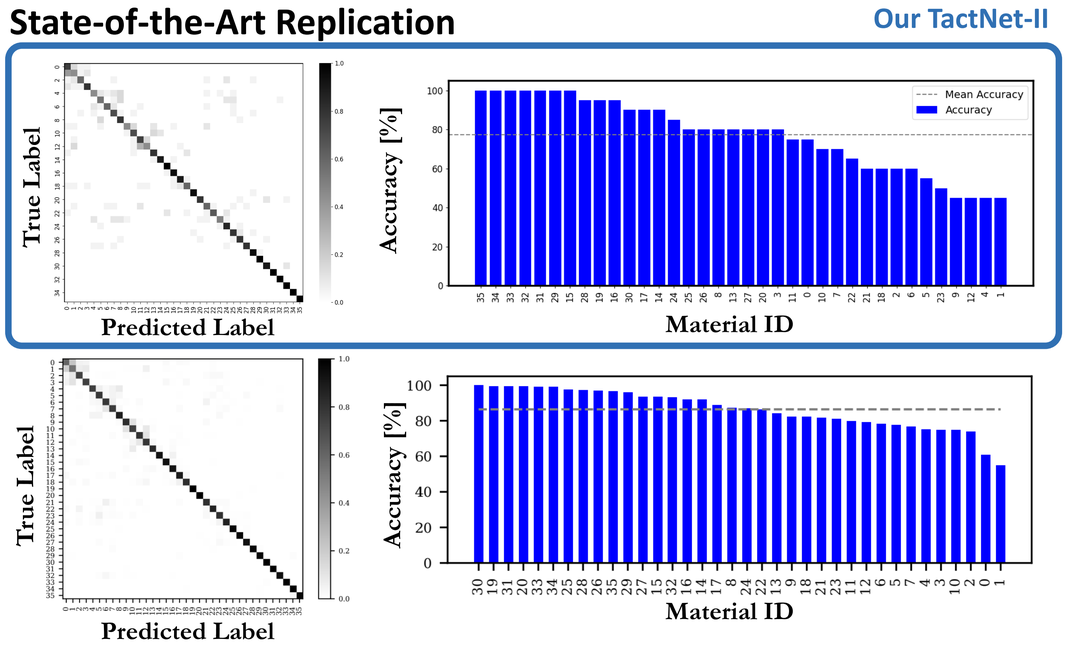


Fig. 2, Confusion Matrix & Class-wise accuracy of *TactNet-II vs, Our Network*

Superhuman performance in tactile classification has been achieved in an experimental environment, reaching up to **86.3% accuracy** in a single-sweep setup [2]. We implemented a simplified version of *TactNet-II* (Fig.3), providing a benchmark model without the Siamese Differentiation network. Instead of targeted adversarial training, we used random noise augmentation to increase classification robustness. Our simplified model achieves **77.2% accuracy**, which, although lower than the benchmark, serves as a solid baseline for evaluating other architectures.

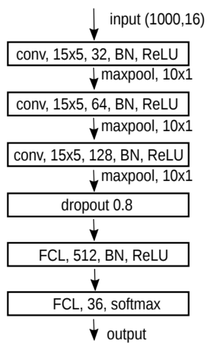


Fig. 3, *TactNet-II* Architecture

TactINN Performance

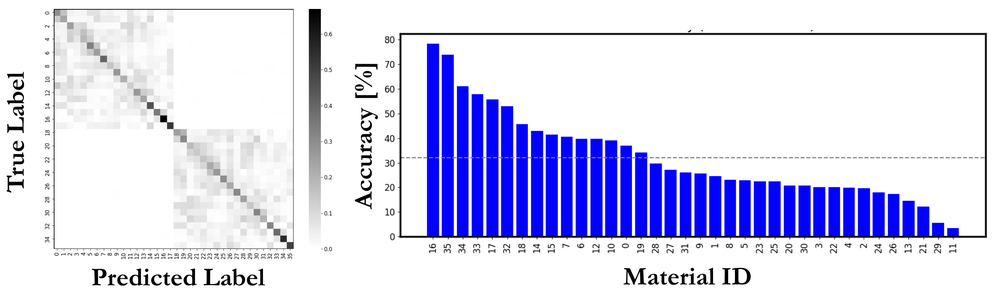


Fig. 4, Confusion Matrix & Class-wise accuracy of *TactINN*

Following the architecture described in [3], we adapt an Invertible Neural Network (INN) that leverages affine coupling blocks and normalizing flows to transform the input samples into a latent space Z modeled as a Gaussian distribution. In this latent space , a Gaussian Mixture Model (GMM) is applied, which has been trained on the latent representations of the material classes to form 36 distinct clusters. To ensure that the latent space represents the underlying data while also prioritizing classification, we employ a combined loss function similar to the original paper.

$$\mathcal{L} = \mathcal{L}_{\text{gen}} + \beta\,\mathcal{L}_{\text{cls}}, \qquad \mathcal{L}_{\text{gen}} = \frac{1}{D}\left[-\log\sum_{k=1}^{K}\exp\left(-\frac{1}{2}d_k\right) - \log|\det J|\right], \qquad \mathcal{L}_{\text{cls}} = \text{CE}\left(-\frac{1}{2}d,\, y\right)$$

The generative component of the loss, L_{gen}, is computed from the log-likelihood of the latent representation by taking the **log-sum-exp** over the distances $\{d_k\}$ ($||z-\mu_k||^2$) to each GMM cluster center, **subtracting** the **log-determinant of the INN's Jacobian** ($\log|detJ|$), and dividing by the latent dimension D. Meanwhile, the classification component L_{cls} is a **cross-entropy loss** computed on the **predicted class probabilities**, where $-\frac{1}{2}d$ acts as the "logits" for each class. The overall loss is dictated by β, which is a hyperparameter that weights the generative and classification objectives (see Table X for hyperparameters). Using this approach, we achieve a classification accuracy of **32.8%** (see Fig.4)**.** This is lower than the current state-of-the-art. The confusion matrix shows a split between the first and second 18 classes, suggesting that the GMM is grouping the data into two main clusters rather than 36, as also seen in Fig. 7. This could be due to the way the data is processed with the Fourier Transformation, or an underlying pattern within the data, that misleads the GMM to cluster it into 2 main clusters.

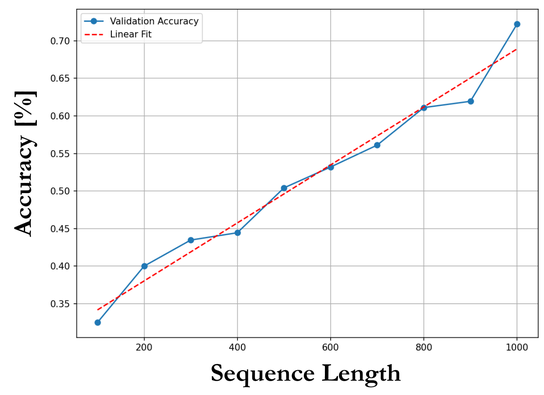
Parameter	Values
Learning Rate	1×10^{-3}
β	3
Batch Size	16
Optimizer	ADAM
n Coupling Blocks	4
Weight decay	$1e^{-4}$
Hidden Layer Size	512

Table 1, Hyperparameters

Sensitivity Study

We conducted a study using 10 different sequence lengths to investigate how sample length affects performance. This is particularly relevant for real robotic systems operating in dynamic environments, where obtaining sufficiently long samples may not always be possible. We adapted the convolutional layers to handle a minimum sequence length of 100, and then incrementally increased the sequence size in steps of 100, which correspond to different contact lengths of the finger. As shown in Fig. 5, there is a linear correlation between sequence length and classification accuracy. We did not observe any specific cutoff at which the accuracy exhibits a notably steeper increase or decrease. We can therefore not derive a singular sequence length, from which we can conclude a sufficiently accurate classification.

Fig. 5, Accuracy vs. Sequence Length



TactINN Workflow

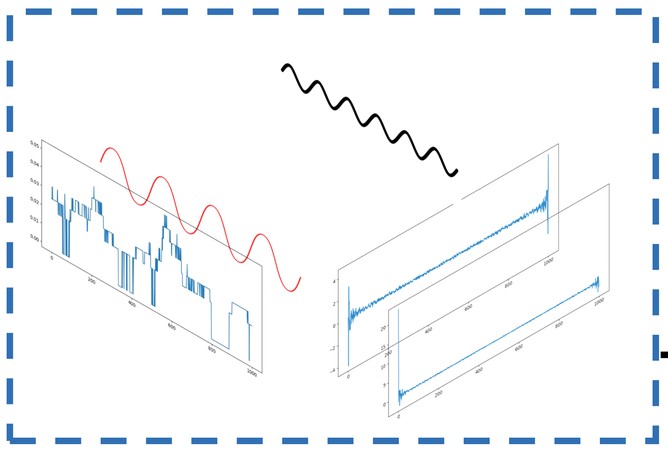


Fig. 6, Fourier Transform of Sample Data

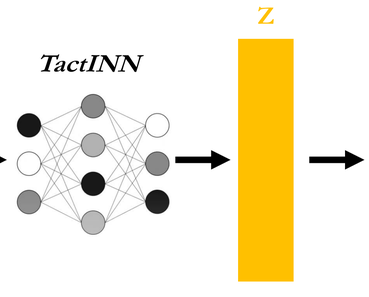
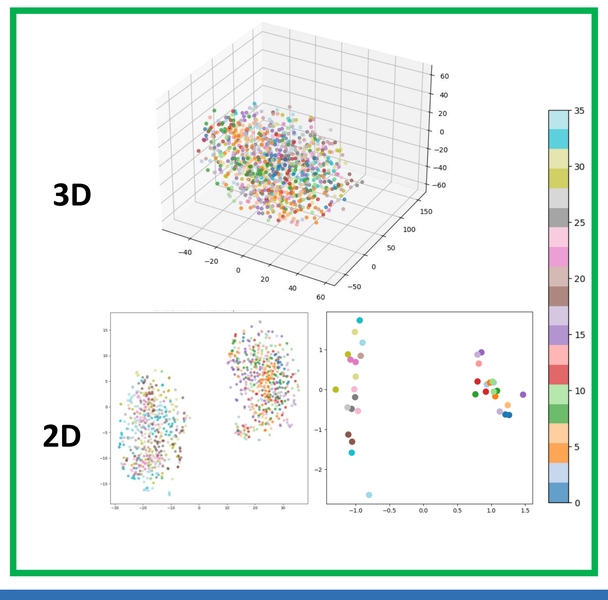


Fig. 7, Low Dimensional (2D/3D) GMM Visualization of the Latent Space (PCA+TSNE Dim. Red. Clustering)



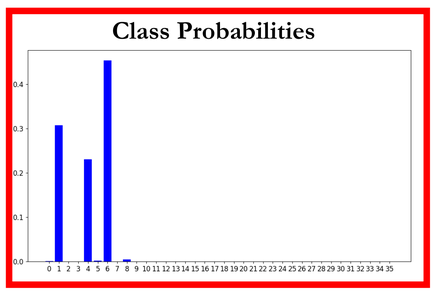


Fig. 8, Prediction Probabilities for the Classes

Sources